TED ZHAO

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PROFILE

- Over 5 years of experience in Machine Learning and Data Science, specializing in recommendation systems and computer vision.
- Experienced in Python backend development, with familiarity in Django, Flask, and SQLAlchemy, for side projects and additional responsibilities.
- Experienced in delivering end-to-end ML solutions using Databricks/Spark, Docker, and Python.

SKILLS

Programming Languages	**Python, VBA, Scala, Julia**
Web/Web Scraping	Django, Flask, asyncio, Scrapy
Cloud	AWS, AWS Sagemaker, Azure ML, Databricks
ETL	SQL, SQLAlchemy, Spark, Pandas
Data Science	Tableau, A/B Testing, Statistics, Data Analysis
AI/Machine Learning	Reinforcement Learning, Deep Learning, Clustering, Lightgbm, XGBoost, SVM, Word2Vec
Version Control	Git
Containerization	Docker

WORK EXPERIENCE

Machine Learning Engineer - Skyscanner

Shenzhen, China (Collaborated with teams across the UK and Bulgaria) *Nov 2019 - July 2021 (1 year and 9 months)*

- **[Recommendation Systems]** Developed and improved recommendation systems using iterative ML models (Logistic Regression / LightGBM / Deep learning).
  - Utilized **Docker** for end-to-end model delivery with **AWS Sagemaker** environment, resulting in a **+5-10% increase in CTR** with each iteration.
- **[Computer Vision / Deep Learning]** Applied Deep learning models to hotel images to enhance user experience.
  - Employed ResNet for categorizing hotel images into 20 landscape/interior labels, achieving **99% accuracy** in Top-5 predictions.
  - Utilized MobileNet to select thumbnails for hotels, resulting in a **+3% increase in CTR**.
- **[A/B Testing]** Performed AB test analysis for car-hire and hotel search pages.
  - Achieved **10X growth** in car-hire during the pandemic.
  - Built data dashboards in Tableau and Databricks for stakeholder reference.
- **[ETL]** Designed and implemented data pipelines in Spark/Databricks.
  - Generated **a dozen key metrics and features** to train models and conduct AB testing analytics.

Data Scientist - Smart Decision Technology Ltd

Shenzhen, China *Aug 2015 - Mar 2018 (2 years and 8 months)*

- **[Simulation]** Designed and implemented a Python-based simulation system for S.F. Express.
  - Leveraged **SimPy** to accurately model the conveyor system within the Distribution Centre.
  - Created an intuitive front-end interface, enabling business users to input site designs and parameters with ease.
  - Developed a robust back-end service using **Flask**, which stored simulation parameters and results for each iteration, utilizing **SQLAlchemy** for efficient model management.
  - The solution surpassed three competitors in terms of speed and adaptability, securing a **2-year contract** with S.F. Express.
- **[Credit Risk Modeling]** Designed credit scorecard workflows using Python to replace SAS.

- Created **feature selection, binning, and WOE calculation** in Python.
- Converted the logistics model into scorecards in Python.

- **[Segment Analysis / NLP]** Delivered client segment reports for Hong Kong Jockey Club.
  - Created segment analysis using clustering and generated data diagrams and charts in reports.
  - Built **sentiment analysis using Word2Vec** on mixed Cantonese and English restaurant reviews.

Data Scientist - DJI

Shenzhen, China *April 2018 - Aug 2018 (5 months)*

- **[Competitive Analysis]** Provided support for a comprehensive campaign through competitive analysis.
  - **[Data Scraping]** Scraped retail location data from competitors with **Scrapy**.
- **[Data Presentation]** Analyzed collected dataset to identify patterns and insights for the campaign team.
  - Translated findings into actionable recommendations.
  - Empowered stakeholders to make informed decisions and optimize the product launch strategy.
- **[Django]** Developed a Django-based API for Jupyter integration extension, enabling seamless SQL querying and data model management.
  - Designed the API to facilitate Jupyter extension integration for streamlined SQL query execution.
  - Maintained data models using Django and SQLAlchemy, ensuring standardized datasets across the platform.

RESEARCH PROJECT

Quantitative Trading with Reinforcement Learning - The University of Adelaide

Adelaide, SA, Australia *Sep 2022 - Present*

- **[Reinforcement Learning]** Conducting a research project while pursuing a master's degree in Machine Learning at The University of Adelaide.
  - Developed a trading strategy for the Crypto market using **Reinforcement Learning**.
  - Designed a reward system with Sharp Ratio, where actions were the next sell/buy quantity of all assets.
  - Created **two Deep Learning modules** to capture price signals and market sentiment.
  - **[Back Testing]** Developed a back-test with metrics **Sharp Ratio, Maximum Drawdown(MDD)** and presented profit curves with plots, considering fees and compound returns.

Education

	Institution	Duration
Master of Machine Learning	The University of Adelaide	June 2021 - June 2023
Bachelor's Degree in Mathematical Sciences	Shenzhen University	September 2010 - July 2014

Certifications

	Date	Verification
Machine Learning - Coursera	October 2016	Verify
Machine Learning Nanodegree Program - Udacity	December 2018	Verify

Language Proficiency

	Proficiency
English	IELTS - CEFR C1 (Advanced)
Mandarin (Chinese)	Native
Cantonese (Chinese)	Native